

MATHEMATICS SUBJECTIVE TEST

Class X (Session 2026 – 27)

Duration : 120 Minutes

Max. Marks : 45

General Instruction :

1. All questions are compulsory.
2. Section A – 5 questions $\times$ 1 mark = 5 marks.
3. Section B – 5 questions $\times$ 3 marks = 15 marks.
4. Section C – 5 questions $\times$ 5 marks = 25 marks.

SECTION – A

(Each question carries 1 marks)

1. Find a and b so that the numbers a, 7, b, 23 are in A.P. (1 mark)
2. Solve the following system of equations for x and y : (1 mark)

$$\frac{x}{2} + \frac{2y}{3} = -1 \text{ and } x - \frac{y}{3} = 3.$$
3. Verify that roots of the quadratic equation $(p - q)x^2 + (q - r)x + (r - p) = 0$ are equal when $q + r = 2p$. (1 mark)
4. Which term of the A.P. : 65, 61, 57, 53, _____ is the first negative term? (1 mark)
5. The sum of the first 15 terms of an A.P. is 750 and its first term is 15. Find its 20th term (1 mark)

SECTION - B

(Each question carries 3 marks)

6. In an A.P., 15th term exceeds the 8th term by 21. If sum of first 10 terms is 55, then form the A.P. (3 mark)
7. Prove that $2 + 3\sqrt{5}$ is an irrational number given that $\sqrt{5}$ is irrational. (3 mark)
8. Find the zeroes of the quadratic polynomial $x^2 - 15$ and verify the relationship between the zeroes and the coefficients of the polynomial. (3 mark)
9. The third and ninth term of an A.P. are 4 and -8 respectively. (3 mark)
 - (i) Which term of the A.P. is zero?
 - (ii) Find the value of n is $S_n = -36$
10. The minimum age of children eligible to participate in a painting competition is 8 years. It is observed that the age of the youngest boy was 8 years and the ages of the participants, when seated in order of age, have a common difference of 4 months. If the sum of the ages of all the participants is 168 years, find the age of the eldest participant in the painting competition. (3 mark)

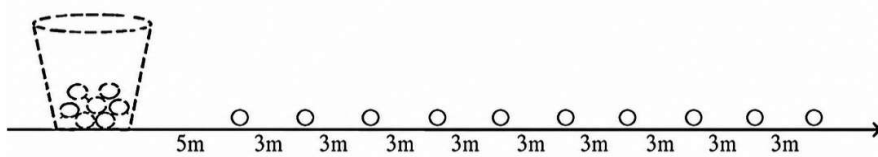
SECTION - C

(Each question carries 5 marks)

11. The area of a right angled triangle is 600 cm². If the base of the triangle exceeds the altitude by 10 cm, find all the three dimensions of the triangle. (5 mark)
12. A fraction becomes $\frac{5}{6}$ when 3 is added to both the numerator and the denominator. If 2 is added to both the numerator and the denominator, the fraction become $\frac{9}{11}$. Express the given information algebraically as a system of linear equations in two variables. Hence, find the original fraction. (5 mark)

13. In a potato race, a bucket is placed at the starting point, which is 5 m from the first potato. The other potatoes are arranged 3 m apart in a straight line, with a total of 10 potatoes, as shown in the figure.

(5 mark)

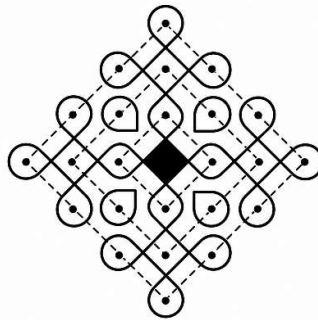


OR

A competitor starts from the bucket, picks up the nearest potato, runs back to the bucket to drop it in, then returns to pick up the next potato. This process continues until all the potatoes are in the bucket.

Based on the above information, answer the following questions :

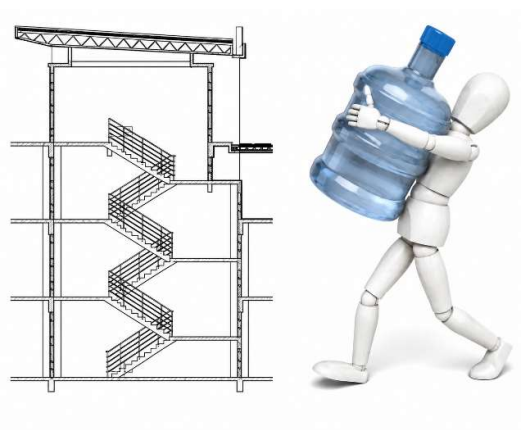
- (i) What is the distance covered to pick up the first potato and drop it in bucket?
 - (ii) What is the distance covered to pick up the second potato and drop it in bucket?
 - (iii) (a) What is the total distance the competitor has to run?
 (b) If average speed of competitor is 5 m/s, then find the average time taken by competitor to put all the potatoes in the bucket.
14. 'Kolam' is a decorative art which is made with rice flour in South Indian States. It is drawn on grid pattern of dots. One such art work is shown below. (5 mark)



OR

Observe the given figure carefully. There are 4 dots in first square, 8 dots in second square, 12 dots in third Square and so on.

- (i) Show that number of dots given above form an A.P. Write the first term and common difference.
 - (ii) Write n^{th} term of the A.P. formed.
 - (iii) The pattern is expanded on a large ground. If total 220 dots are used, then find the number of squares formed.
15. A multistorey building is constructed with stilt parking. There is a provision of lift, as well as staircase from the ground floor to the top floor. The number of stairs from the ground floor to the first floor is 10, from the first floor to the second floor is 24, from the second floor to the third floor is 38, and so on. (5 mark)



Based on the above information, answer the following questions :

- (i) Does 10, 24, 38, ... form an A.P. ? Justify your answer.
- (ii) What will be the total number of stairs from the ground floor to the eleventh floor?
- (iii) (a) A person supplies water cans to people living in the building. As water cans are heavy, he supplies water cans on each floor, carrying one at a time. He supplied the water can from the ground floor to the first floor, came back and supplied water can to the second floor, again came back then supplied water can to the third floor, and so on. Find the total number of stairs he climbed up and down to supply water till the sixth floor, using A.P.

OR

- (iii) (b) The next day, following the same process, if a person climbed up and down a total of 380 stairs, till which floor did he supply water cans?

